

Transition-risk core: how L , Δp , and the GVA shock

- A — technical coefficients (global MRIO)
- L — the Leontief inverse
- Δp — the carbon-tax price change, per sector
- $\Delta V/V$ — the relative GVA (value-added) shock, per sector → the paper's Table 4

1. Inputs

Symbol	Meaning	Source
Z	inter-industry flow matrix (1000×1000)	DATA_20R/ICIO02025_20R_2022.csv , region-industry block
x	gross output per sector (1000)	ICIO OUT row
GVA	value added per sector (1000)	ICIO VA row
CI	direct carbon intensity, tCO ₂ e per USD-million output	DATA_20R/CARBON_INTENSITY_20R_2022.csv
XCE	carbon price, USD/tCO ₂ e	parameter = 70 (paper's Table 4 value)
ϕ	cost pass-through fraction	parameter, swept 0 → 1 in 10% steps

The index is 20 regions × 50 ISIC industries = **1000 region-sectors** (labels like GBR_C24A). ROW is the accounting closure region and is kept in the matrix so the global IO system balances, but carries no financial interpretation.

Units. ICIO is in current USD millions; CI is tCO₂e per USD-million of output. With XCE in USD/t, the per-output carbon charge is a **dimensionless fraction of output value**:

$$ct_j = CI_j \times XCE \times 10^{-6}.$$

The 10^{-6} reconciles the units: the charge $CT_j = CI_j x_j \times XCE$ comes out in plain USD, while output x_j is in USD *millions*, so dividing the two requires the factor of a million.

2. Step 1 — technical coefficients A and Leontief inverse L

Column-normalise the flow matrix by gross output (Eq. 2), then invert:

$$A = Z \hat{x}^{-1}, \quad A_{ij} = \frac{Z_{ij}}{x_j}, \quad L = (I - A)^{-1}.$$

Zero-output columns are guarded before the division.

$\tilde{L}$ is the **demand-side** Leontief inverse; its column sums are sector output multipliers. Because ICIO is a true multi-regional table, A is the full global block matrix, so $\tilde{L}$ embeds cross-border linkages automatically (a shock in one region propagates to all others via the off-diagonal trade blocks).

The price side uses the **transpose (dual)** form $\tilde{L} = (I - A^T)^{-1}$ (Eq. 4).

A validity check: the spectral radius of A must be < 1 (productive economy). For this data it is ≈ 0.587 , so $(I - A)$ is invertible.

3. Step 2 — carbon charge vector ct_direct

Multiply each sector's carbon intensity by the carbon price (Eq. 6), aligning the carbon-intensity table (industries $\times$ regions) to the ICIO region-sector ordering:

$$ct_{direct} = CI \times XCE \times 10^{-6}.$$

This is the fraction of each sector's output value taken by the carbon tax on its **Scope-1** emissions. (Scope 2/3 are captured up/downstream through the IO linkages, so only direct emissions enter here.)

4. Step 3 — price change Δp (modified Leontief dual, Eq. 8)

Only a fraction ϕ of the charge is passed into prices. With $\hat{\phi} = \text{diag}(\phi)$, the **modified Leontief dual** and the resulting price change are:

$$\tilde{L}(\phi) = (I - A^T \hat{\phi})^{-1} \hat{\phi}, \quad \Delta p = \tilde{L}(\phi) ct_{direct}.$$

Δp_j is the relative increase in sector j 's output price. It is **0 at $\phi=0$** (nothing passed on) and rises monotonically to full pass-through at $\phi=1$.

5. Step 4 — GVA shock $\Delta v/v$ (Eq. 10)

The carbon charge splits into a *price* part (Δp) and a *retained-in-GVA* part. Differencing the sector cost identity before/after the tax (Eqs. 3, 5, 9) under inelastic demand (final demand y fixed) gives the per-unit-output value-added change:

$$\Delta v = \left[(I - A^T) \tilde{L}(\phi) - I + \hat{\phi} \right] ct_{direct}.$$

Scale to absolute and then to a **relative** shock:

$$\Delta V_j = x_j \Delta v_j, \quad \boxed{\frac{\Delta V_j}{V_j} = \frac{x_j \Delta v_j}{GVA_j}}.$$

Sign convention. Negative = value added lost. Δp and the GVA shock are complementary: cost not passed into prices shows up as a GVA loss.

Sanity anchors (the paper's two special cases)

φ	Δp	GVA shock $\Delta v/v$
0 (absorb all)	0	$-CT_j/GVA_j$ (most negative)
1 (pass all on)	max	$+CT_j/GVA_j$ (mirror image)

The script verifies the region-level version: UK shock is **-0.851%** at $\varphi=0$ and **+0.851%** at $\varphi=1$, matching $\mp CT_{UK}/GDP_{UK}$.

6. Step 5 — aggregating per-sector → per-nation

A region's GDP is the sum of its sectors' GVA, so the national relative shock is the total ΔGVA over total GVA — equivalently a **GDP-share-weighted average** of the per-sector shocks:

$$\frac{\Delta GDP_r}{GDP_r} = \frac{\sum_{j \in r} \Delta V_j}{\sum_{j \in r} GVA_j} = \sum_{j \in r} f_{j,r} \frac{\Delta V_j}{V_j}, \quad f_{j,r} = \frac{GVA_j}{\sum_{k \in r} GVA_k}.$$

The weighting scheme differs by quantity: the GVA/GDP shock is weighted by **GVA share**, whereas Δp is rolled up with **output weights**, since a producer-price index is output-weighted rather than value-added-weighted.

The per-sector Δv already contains cross-border cascading, so a national number includes the drag from pricier imported inputs; aggregation just sums the region's own sectors.

7. Outputs

File	Content
out_gva_shock_by_region_phi.csv	national GVA/GDP shock (%), 20 regions $\times$ φ
out_price_change_by_region_phi.csv	national output-weighted Δp (%), 20 regions $\times$ φ
out_gva_shock_by_sector_phi.csv	full 1000 region-sectors $\times$ φ GVA shock (%)
out_price_change_by_sector_phi.csv	full 1000 region-sectors $\times$ φ Δp (%)